

Project Goal

To develop a comfortable and user-friendly wearable device that provides real-time data analysis of movements of the knee joint. Athletes could use the device for monitoring and training to reduce their risk of a knee injury.

Background

- Anterior Cruciate Ligament (ACL) injuries are a common sport injury and occur when the ligament of an individual is compromised due to compressive loading, valgus bending, and/or internal rotation of the tibia.

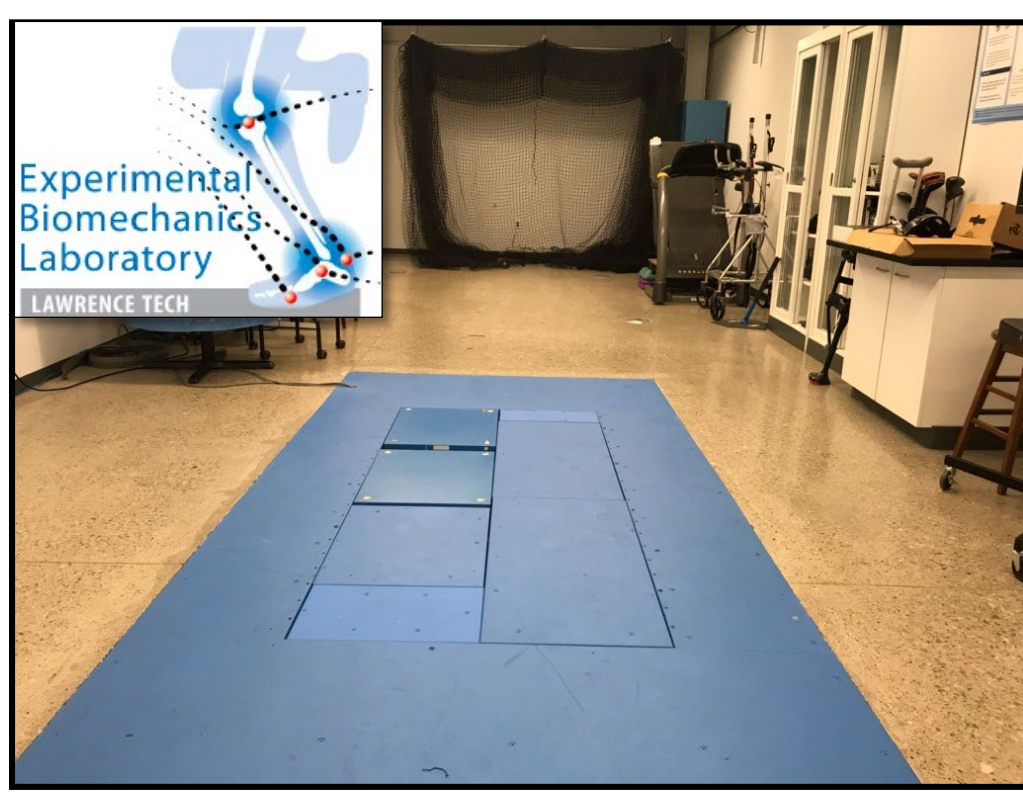


Figure 2 – Vicon Motion Capture System in Biomechanics Lab at Lawrence Tech. University. A monitoring system that uses infrared cameras and force plates to gather dynamic angle measurements.

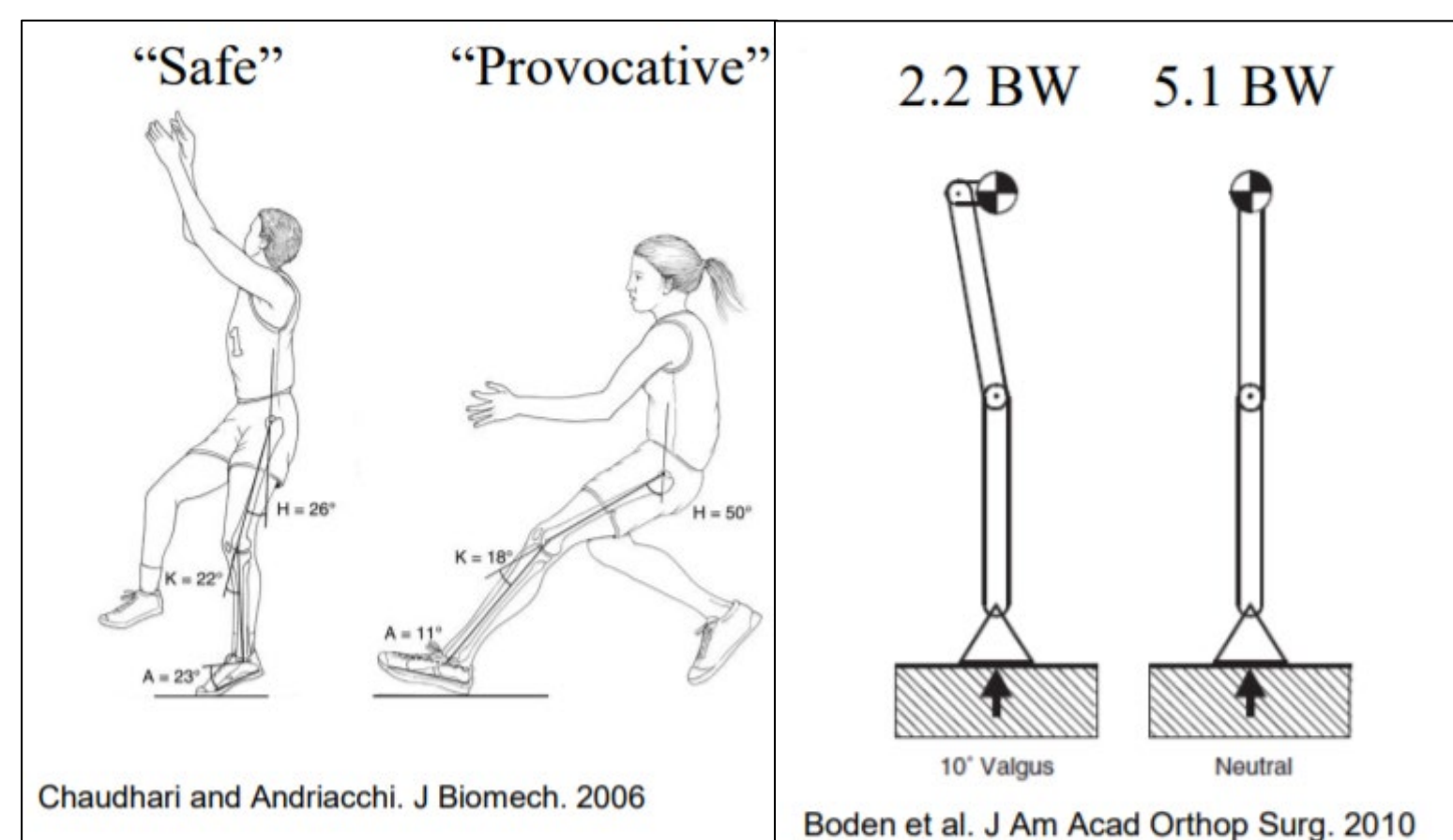


Figure 1 – Sagittal and Frontal Plane of Injury Mechanism.

- 100,000 to 250,000 ACL tears are reported annually in the United States. About half of these injuries require reconstructive surgery.
- Current monitoring systems are expensive and/or immobile.

Device Design

- A mesh sleeve was used for user comfort and to allow full range of motion of the knee.
- Two Adafruit BNO055 Absolute Orientation Sensors are placed on the sleeve, above and below the knee.
- Wiring for the device is integrated in the sleeve to keep the device unobtrusive.



Figure 3 – Sleeve Design.

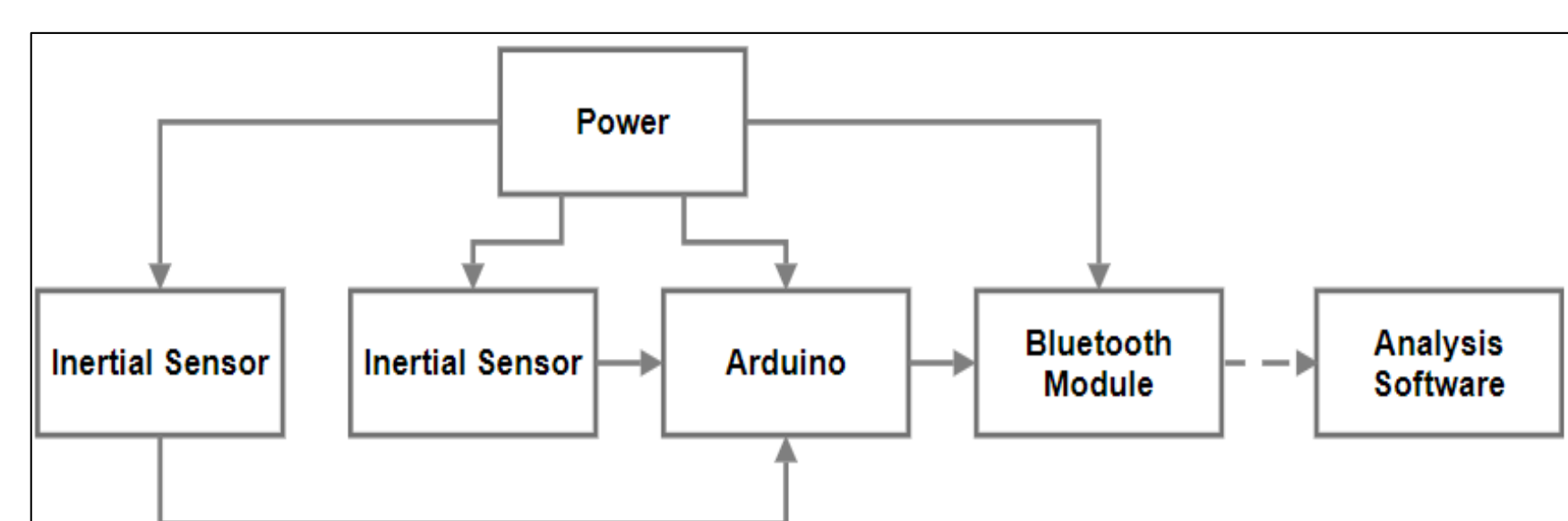


Figure 4 – System Block Diagram.

- Each inertial sensor includes an accelerometer, gyroscope, and magnetometer, outputting 3D acceleration, 3D angular velocity, and 3D orientation relative to the ground.

- An Arduino microprocessor collects the data and gives an output of relative angle and acceleration.
- 3D printed sensor and microprocessor cases protect the device components from water and impact damage.
- The cases are also integrated into the sleeve to hold them in place.

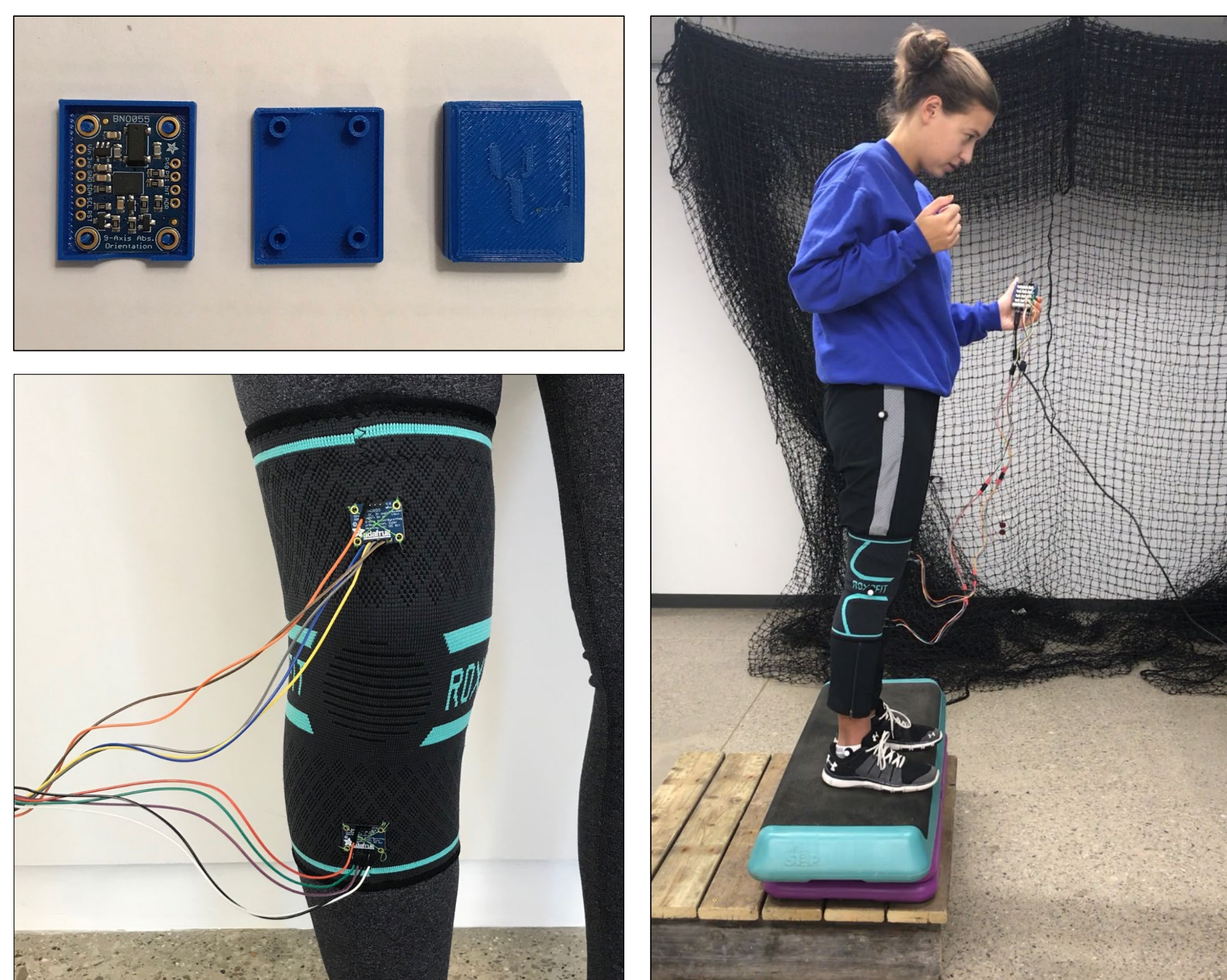


Figure 5 – 3D printed sensor case (top left). Prototype Model – Two Adafruit Absolute Orientation Sensors were placed on a prefabricated sleeve and worn by a human subject for testing (bottom left). Subject performing drop landing during validation testing (right).

Angle Calculation

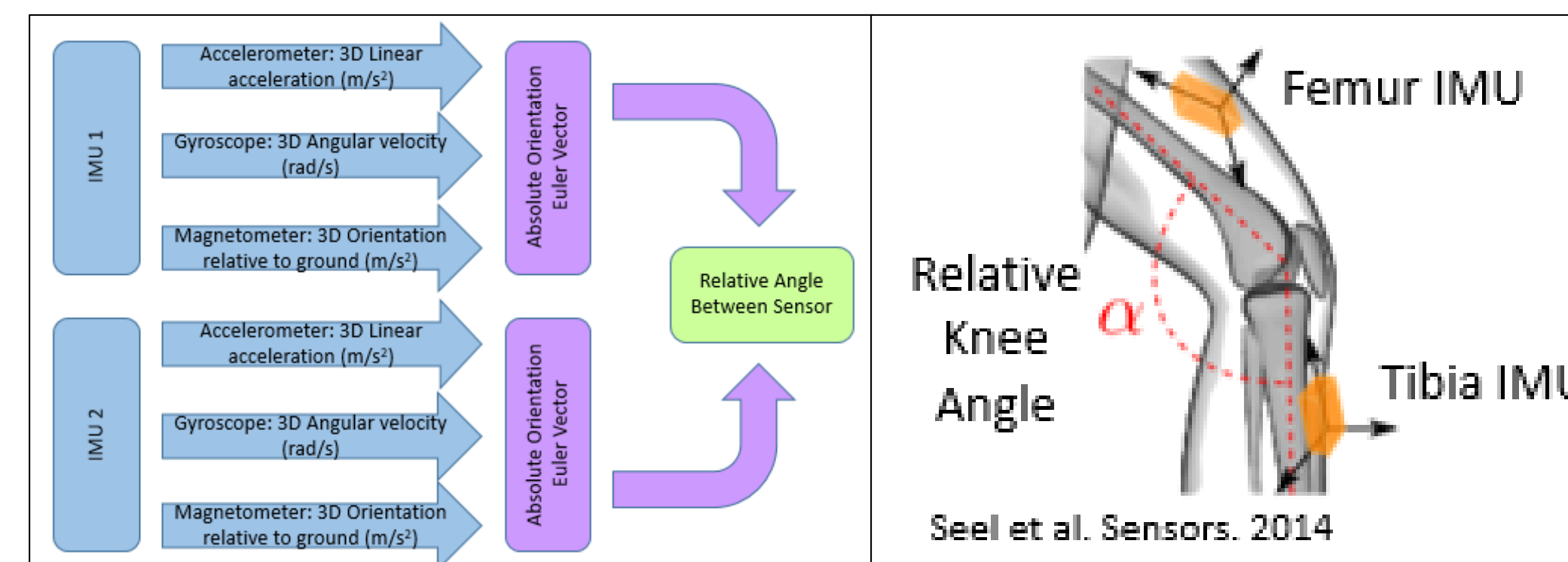


Figure 6 – IMU based Angle Calculation (left) and IMU Angle Diagram (right).

- Sensor fusion is used to calculate an absolute orientation Euler vector for each sensor.
- Euler vectors are compared to give a relative angle between the sensors.

Testing and Results

- Preliminary testing was performed by having a subject wear the prototype and perform a defined motion. The subject stood up straight and raised their foot off the ground, bending the knee, to mimic the natural full range of motion of the knee, 0-135 degrees.

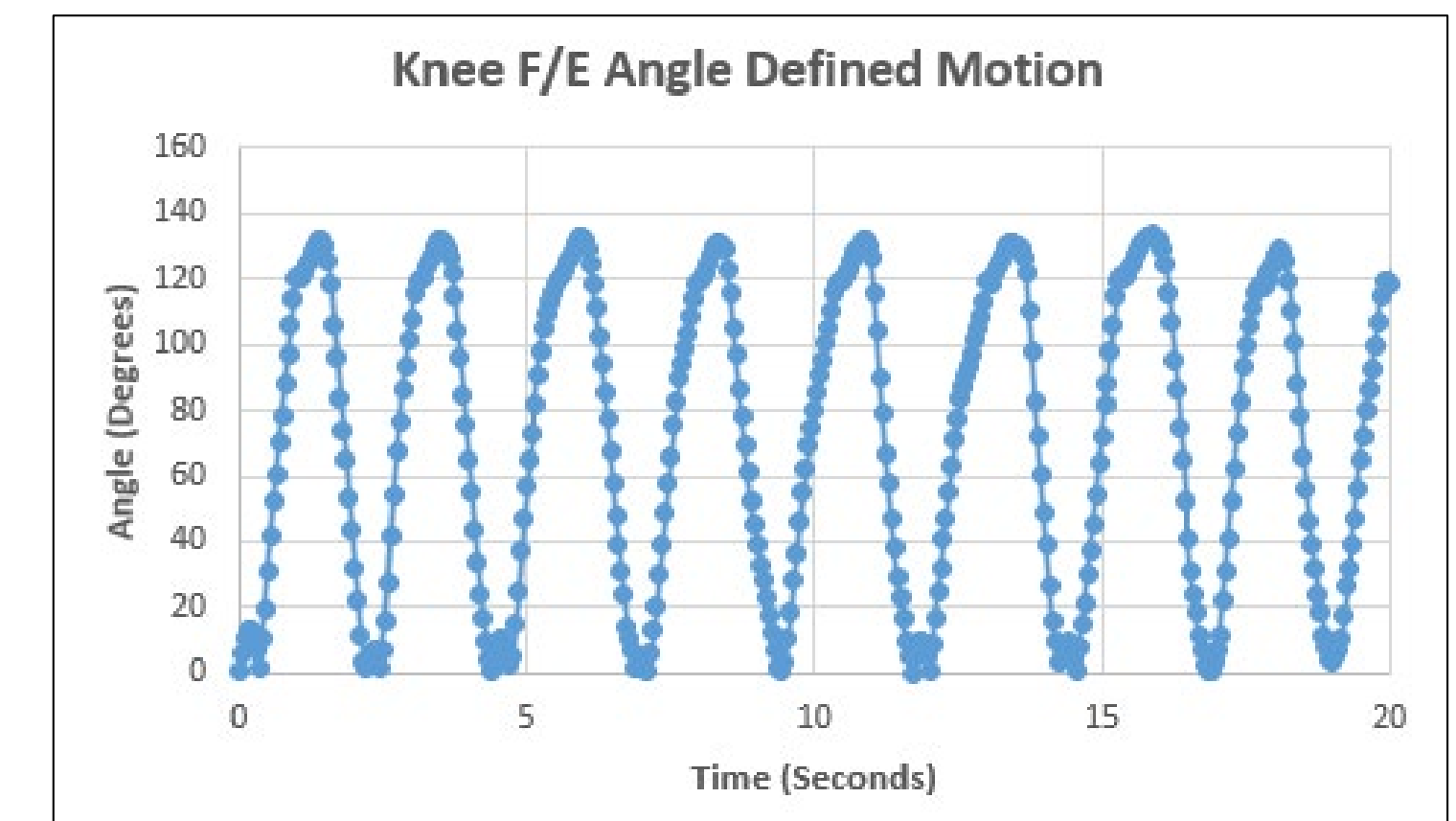


Figure 7 – Knee Flexion/Extension Angle Performing Defined Motion.

- Validation testing was performed by having a subject wear the prototype while performing a standing squat and a drop landing. Kinovea video analysis software was used to compare device results with video results.

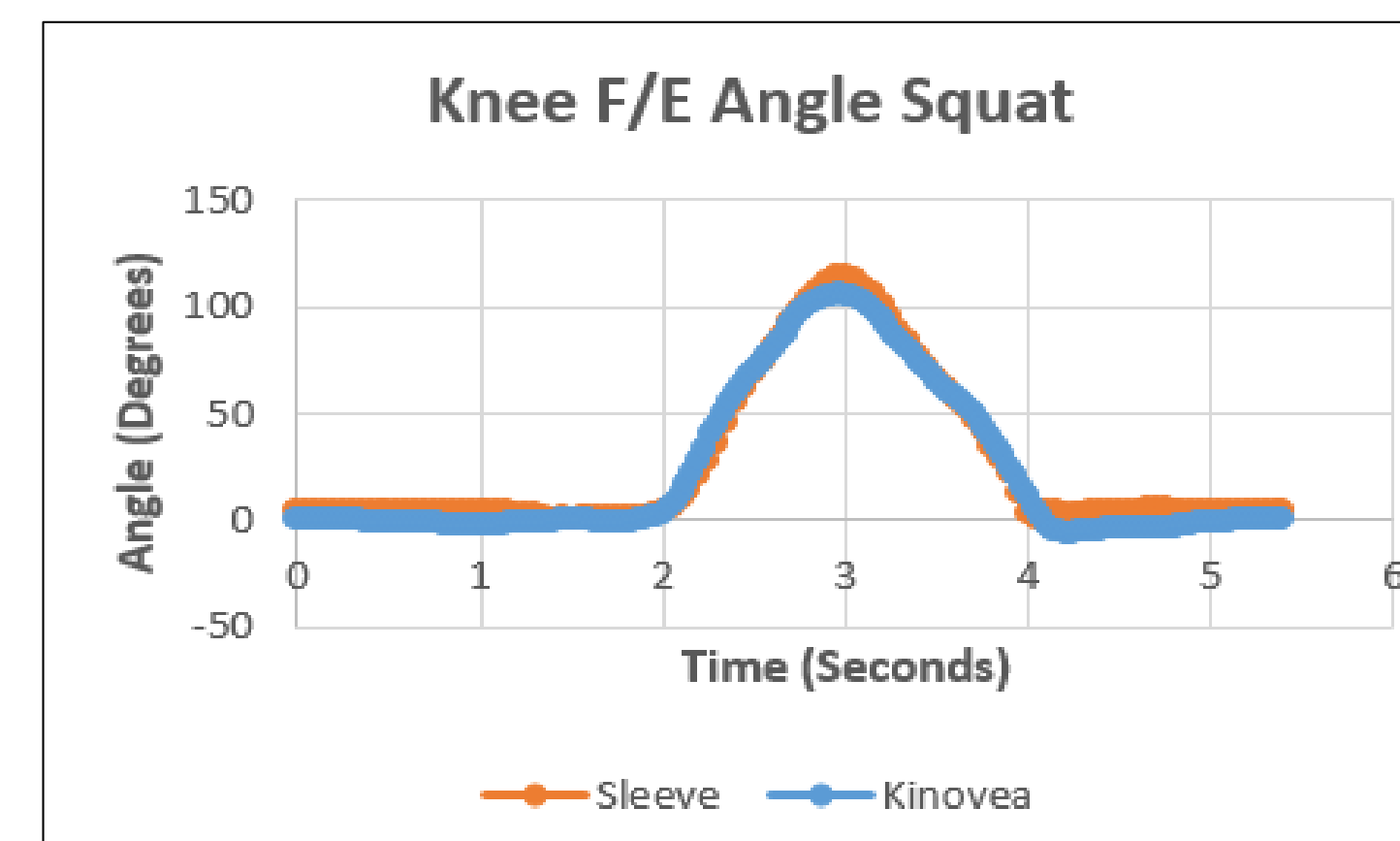


Figure 8 – Knee Flexion/Extension Angle during Standing Squat.

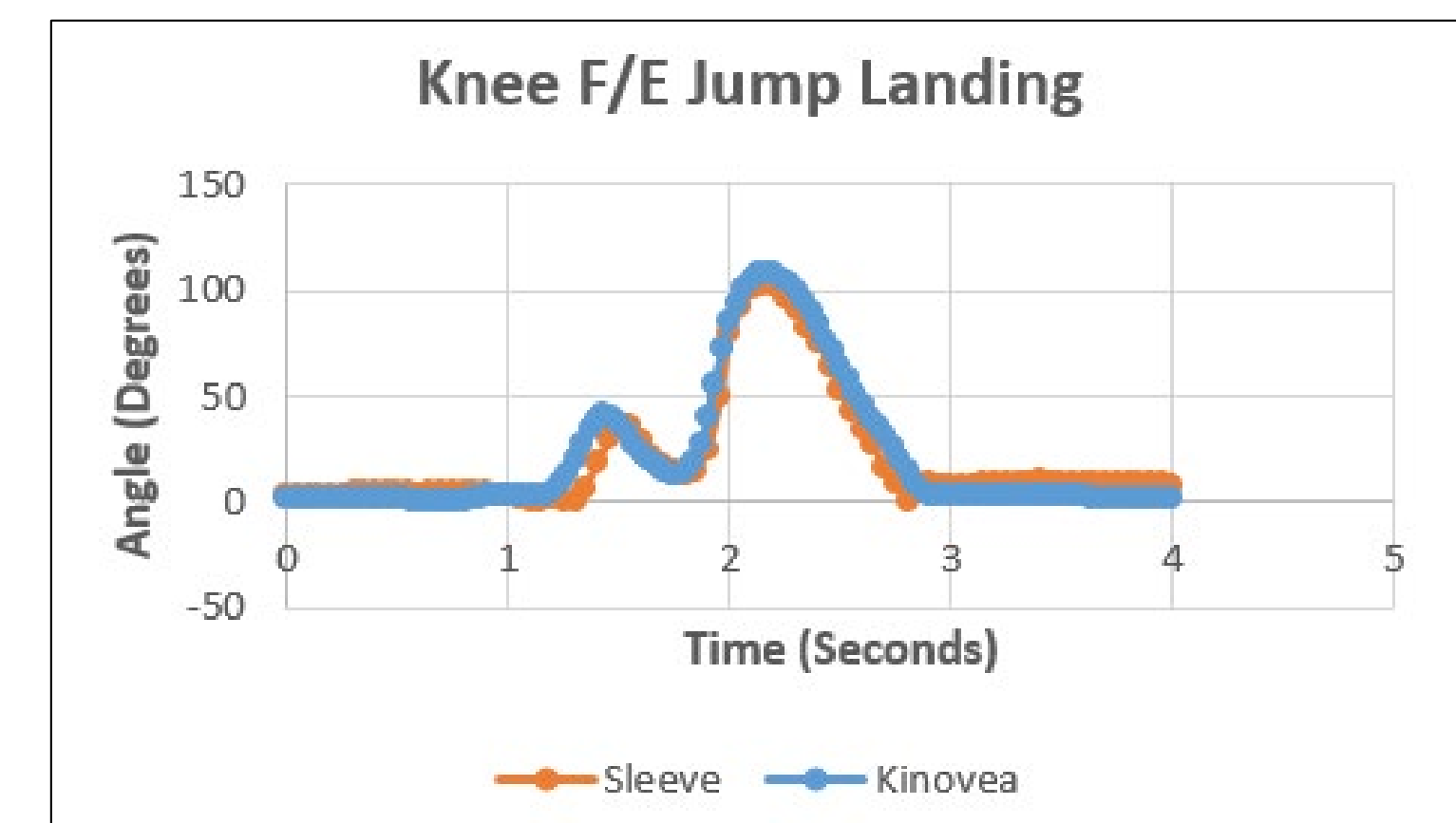


Figure 9 – Knee Flexion/Extension Angle during Drop Landing.

- Tibial acceleration was also recorded for a standing squat and jump landing while wearing the device.
- For the standing squat, there is no tibial acceleration because the tibia is not moving. For the Jump landing, the peak tibial acceleration occurs at 38 m/s².

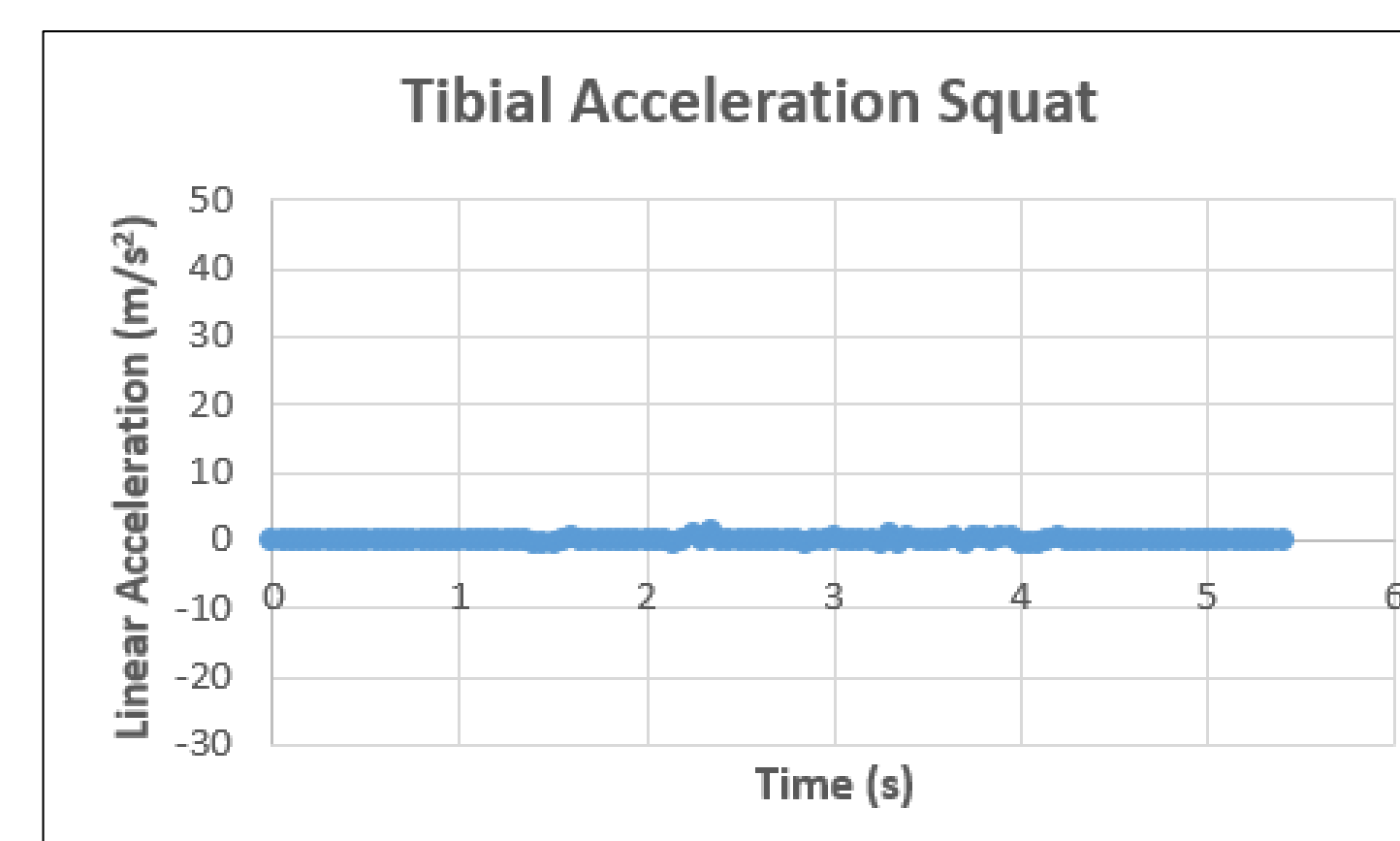


Figure 10 – Tibial Acceleration during Standing Squat.

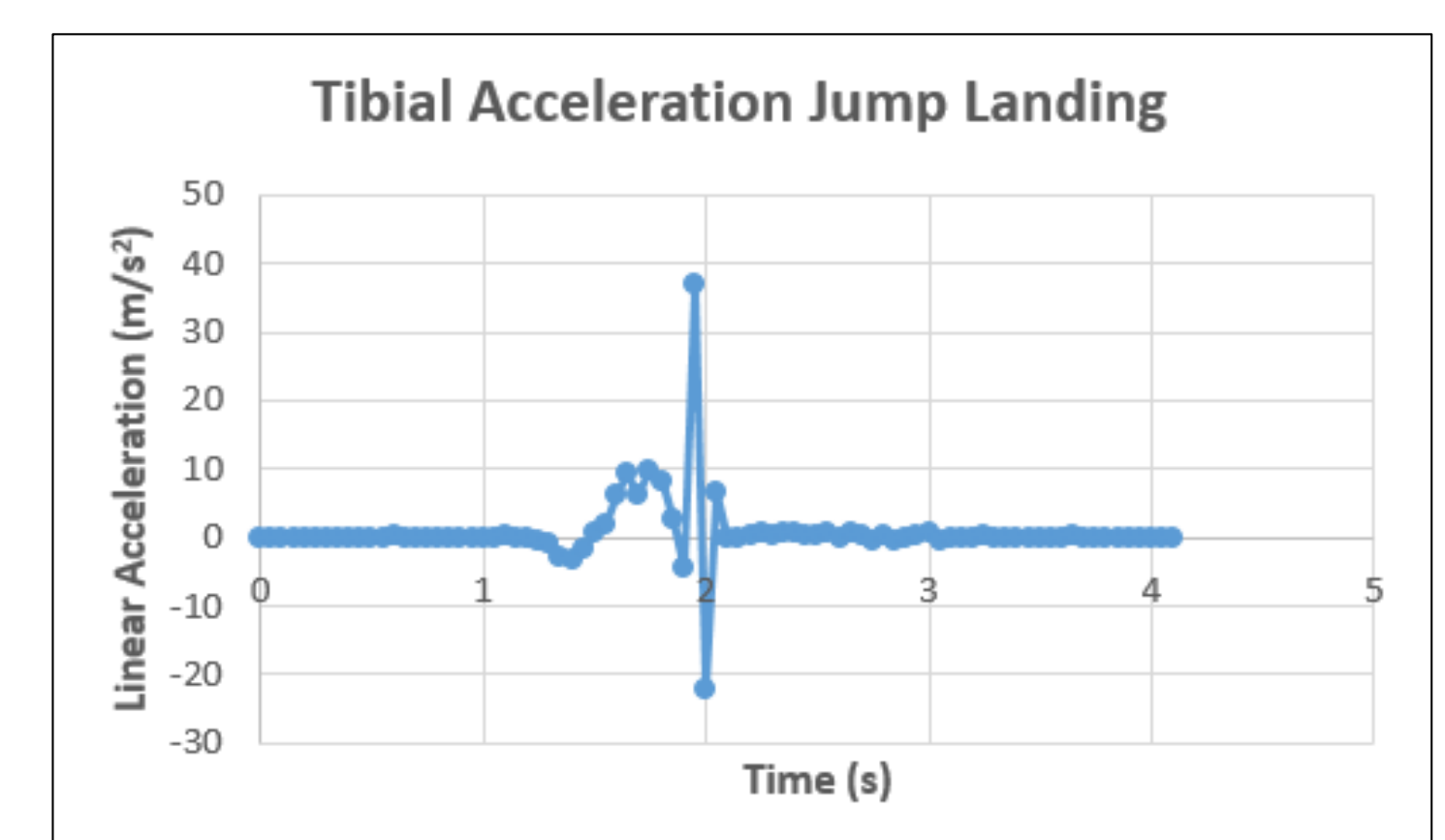


Figure 11 – Tibial Acceleration during Drop Landing.

Conclusion

- Using the tibial acceleration values, we can find when the foot strikes the ground.
- Upon foot strike, the knee flexion angle can be recorded, along with a maximum knee flexion angle. These values can be used to calculate the knee flexion range of motion.

Future Work

- Additional parametric thresholds for added applicability and accuracy.
- Bluetooth/wireless enabled for free range monitoring.
- Integrated desktop/smartphone app for easy visualization of data outputs.

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